

MOF_benchmark: Module A Documentation

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July 23, 2026

Abstract

This document describes the current state of `MOF_benchmark` up to the first runnable version of Module A. Module A targets single-component CO₂ adsorption in a rigid MOF framework using GCMC-style LAMMPS input generation. The current implementation can resolve benchmark inputs, generate LAMMPS input artifacts, execute pressure-point runs, parse LAMMPS logs, and evaluate the resulting adsorption isotherm against curated reference data. The simulation output is treated as absolute adsorption from molecule counts, and the evaluation layer can additionally calculate excess adsorption when a pore volume is available.

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1 Project Goal

The goal of `MOF_benchmark` is to provide a reproducible benchmark pipeline for gas adsorption simulations in metal-organic frameworks. The project is structured around benchmark modules:

- Module A: single-component adsorption in a rigid framework.
- Module B: mixture adsorption and selectivity.
- Module C: Widom insertion, Henry coefficients, or force-field comparison.
- Module D: flexible-framework simulations or MD/GCMC flexibility studies.

The current implementation focuses on Module A, using MOF-5/IRMOF-1 and CO₂ as the first benchmark case.

2 Current Scope of Module A

Module A currently covers the following workflow:

1. Load a benchmark JSON input file.
2. Normalize missing configuration values with defaults.
3. Resolve the material alias MOF-5 to IRMOF-1.
4. Resolve the CIF file and charge scheme.
5. Resolve force-field files and adsorbate definitions from CRAFTED.
6. Select the benchmark module automatically.
7. Build a side-effect-free run plan.
8. Materialize a run directory with generated LAMMPS data, molecule templates, force-field include files, and GCMC inputs.
9. Run a short smoke test or a full pressure-grid isotherm.
10. Parse LAMMPS logs into JSON summaries.
11. Evaluate the isotherm against CRAFTED and NIST reference data.
12. Report both absolute and excess adsorption where possible.

3 Benchmark Configuration

The main Module A input is stored in:

`input_json_files/benchmark_tao2022_mof5_co2.json`

The configuration contains the following conceptual sections:

- **material**: framework name, CIF source, charge scheme, and data source.
- **adsorbates**: adsorbate components, currently CO₂.
- **conditions**: temperature and pressure grid.

- **benchmark:** task, metrics, and reference-data configuration.
- **simulation:** LAMMPS/GCMC settings.
- **evaluation:** absolute/excess handling and gas-density model.
- **output:** output directory, run ID, overwrite behavior, and saved artifacts.

3.1 Current Paper-Like Setup

The current paper-like MOF-5/CO2 setup uses:

Setting	Current value
Material	MOF-5, resolved as IRMOF-1
Charge scheme	DDEC
Adsorbate	CO2
Temperature	298.15 K
Pressure points	0.1, 0.5, 1, 2, 5, 10, 20, 35 bar
Force field	UFF framework and CRAFTED CO2 definition
Framework model	Rigid
Unit cells	1 x 1 x 1
LAMMPS units	real
Pair style	lj/cut/coul/long
Cutoff	12.8 Angstrom
KSpace	pppm, accuracy 1e-4
Fugacity model	Peng-Robinson via CoolProp
Production cycles	500000
Initialization cycles	500000

4 Pipeline Design

4.1 Planning

The planning stage is side-effect-free. It creates a JSON-serializable run plan without writing output files or starting LAMMPS. The main command is:

```
python benchmark.py benchmark_tao2022_mof5_co2.json --dry-run
```

The run plan contains:

- selected module,
- resolved material and CIF path,
- resolved charge scheme,
- resolved force-field files,
- adsorbate definitions,
- pressure and temperature points,
- reference candidates,
- output paths,
- evaluation settings.

4.2 Preparation

The preparation stage creates a side-effect-free description of all files that should be generated. The command is:

```
python benchmark.py benchmark_tao2022_mof5_co2.json --prepare
```

The actual materialization writes:

- framework-only LAMMPS data file,
- CO2 molecule template,
- LAMMPS force-field include file,
- one run-zero input,
- one GCMC smoke-test input,
- one GCMC input per pressure point,
- source copies of CIF, force-field, molecule, and reference files.

5 Converters

5.1 CIF to LAMMPS Data

The CIF converter reads the framework geometry with ASE and separately parses CIF charges from `_atom_site_charge`. This is necessary because ASE reads the geometry and cell but does not reliably preserve the charge column as required for LAMMPS `atom_style full`.

The generated framework data file contains:

- atom count,
- atom types,
- triclinic box parameters,
- masses,
- `Atoms # full` section with molecule ID, atom type, charge, and Cartesian coordinates.

For IRMOF-1 the converter generates a restricted triclinic LAMMPS box. This preserves the periodic cell geometry in the LAMMPS representation.

5.2 Molecule Definition to LAMMPS Template

The molecule converter parses CRAFTED/RASPA `.def` files for adsorbates such as CO2 and N2. For CO2 it creates a LAMMPS molecule template with:

- three atoms,
- two rigid bonds,
- global atom type IDs,
- partial charges from the pseudo-atom definitions.

Because the CO2 template contains bonds, LAMMPS input generation reserves additional per-atom bond and special-neighbor slots using options such as:

```
extra/bond/per/atom 2 extra/special/per/atom 2
```

5.3 Force Field to LAMMPS Include

The force-field converter parses CRAFTED UFF files and writes LAMMPS pair coefficients. It assigns atom types for both framework atoms and adsorbate atoms, then applies Lorentz-Berthelot mixing for cross interactions.

6 LAMMPS Execution

6.1 Smoke Test

A short test run can be started with:

```
conda activate MOF_sim
python benchmark.py benchmark_tao2022_mof5_co2.json --run-test
```

The smoke test verifies that LAMMPS can read the generated data, molecule template, and pair coefficients, and that GCMC insertion can occur.

6.2 Full Isotherm

A full pressure-grid run can be started with:

```
conda activate MOF_sim
python benchmark.py benchmark_tao2022_mof5_co2.json --run-isotherm --new-run --jobs 4
```

The `-new-run` flag creates a timestamped output directory to avoid overwriting older results. The `-jobs` flag controls how many pressure points are executed in parallel.

7 Evaluation

7.1 Absolute Adsorption

The raw LAMMPS output is converted to an absolute loading using the mean number of adsorbate molecules in the simulation cell:

$$n_{\text{abs}} = \frac{1000 \cdot \langle N_{\text{ads}} \rangle}{m_{\text{framework}}} \quad (1)$$

where:

- n_{abs} is the absolute adsorption loading in mol kg^{-1} ,
- $\langle N_{\text{ads}} \rangle$ is the mean number of adsorbate molecules per simulation cell,
- $m_{\text{framework}}$ is the framework mass in atomic mass units.

The numerical values of mol kg^{-1} and mmol g^{-1} are identical.

7.2 Excess Adsorption

The evaluation layer can additionally calculate excess adsorption:

$$n_{\text{ex}} = n_{\text{abs}} - \rho_{\text{gas}} V_{\text{pore}} \quad (2)$$

where:

- n_{ex} is the excess adsorption loading in mol kg^{-1} ,

- n_{abs} is the absolute adsorption loading in mol kg^{-1} ,
- ρ_{gas} is the bulk gas molar density in mol m^{-3} ,
- V_{pore} is the pore volume in $\text{m}^3 \text{kg}^{-1}$.

For IRMOF-1, the current automatic pore volume source is:

CRAFTED-2.0.0/RAC_DBSCAN/CRAFTED_MOF_geometric.csv:AV_cm^3/g

with:

$$V_{\text{pore}} = 0.845545 \text{ cm}^3 \text{ g}^{-1} = 0.000845545 \text{ m}^3 \text{ kg}^{-1}. \quad (3)$$

The gas density is calculated through CoolProp using the configured backend. For production evaluation, the current backend is:

HEOS

7.3 Reference Matching

The evaluation distinguishes reference adsorption bases:

- **simulation_reference**: compared against absolute simulation loading.
- **absolute**: compared against absolute simulation loading.
- **excess**: compared against excess simulation loading.
- **unknown**: reported as unknown and compared against absolute loading by default.

This behavior is controlled by:

"reference_matching": "by_basis"

7.4 Evaluation Outputs

For each evaluated run, the pipeline writes:

- `evaluated_isotherm.csv`,
- `evaluation_report.md`,
- `isotherm_comparison.png`.

For comparison against all available references, the pipeline writes:

- `evaluations/reference_candidates.json`,
- `evaluations/combined_reference_comparison.csv`,
- `evaluations/combined_reference_comparison.png`,
- one evaluation subdirectory per active reference.

8 Reference Data

The current workflow supports two reference sources:

- CRAFTED is used as the primary simulation-reference dataset.
- NIST ISODB is used as an additional experimental/literature reference source.

NIST entries can be excluded from evaluation if they are detected as outliers or if their adsorption basis is not allowed by the benchmark configuration. Excluded entries are documented in `reference_candidates.json` instead of being silently removed.

9 Visualization

LAMMPS trajectory dumps are written for pressure-point runs when dump output is enabled in the generated GCMC input. The dump files can be opened with the OVITO desktop application. Coloring by particle type is useful to distinguish framework atom types from inserted CO2 atom types.

Typical dump location:

```
outputs/module_A_co2_isotherm/runs/<run_id>/dumps/gcmc_10bar.lammpstrj
```

The number of CO2 molecules changes during GCMC because insertion and deletion moves are part of the grand-canonical ensemble. This is expected behavior and is the basis for calculating adsorption from the mean molecule count.

10 Validation Status

The current test suite checks:

- benchmark JSON loading,
- default configuration normalization,
- MOF-5 to IRMOF-1 alias resolution,
- module selection,
- CRAFTED molecule parsing,
- CIF charge parsing,
- LAMMPS data generation,
- force-field parsing and pair-coefficient generation,
- GCMC input generation,
- log parsing,
- isotherm CSV generation,
- NIST reference parsing,
- combined reference evaluation,
- excess adsorption comparison against excess references.

The latest local test run completed with:

```
Ran 44 tests in 41.431s
OK (skipped=7)
```

The skipped tests depend on optional local tools such as ASE, LAMMPS, or the local NIST ISODB checkout.

11 Current Limitations

Module A is technically runnable, but the scientific validation is not complete yet. The most important remaining limitations are:

- convergence has not yet been quantified,
- uncertainty estimates are not yet calculated,
- multiple independent random seeds are not yet automated,
- block averaging is not yet implemented,
- mean absolute error and mean squared error per reference should be made more explicit in a combined report,
- the best-matching reference should be highlighted automatically.

12 Next Steps

The next development steps for Module A are:

1. Add convergence analysis by plotting loading as a function of simulation step or block index.
2. Add multiple-seed execution for every pressure point.
3. Calculate statistical uncertainty using standard deviation across seeds or block standard error of the mean.
4. Refactor the combined evaluation report to show MAE/MSE per reference.
5. Highlight the best-matching reference dataset.
6. Add a final Module A report that combines input protocol, simulation results, reference comparison, uncertainty, and plots.

13 Example Commands

13.1 Dry Run

```
conda activate MOF_sim
python benchmark.py benchmark_tao2022_mof5_co2.json --dry-run
```

13.2 Prepare Files

```
conda activate MOF_sim
python benchmark.py benchmark_tao2022_mof5_co2.json --prepare --new-run
```

13.3 Run Full Isotherm

```
conda activate MOF_sim
python benchmark.py benchmark_tao2022_mof5_co2.json --run-isotherm --new-run --jobs 4
```

13.4 Evaluate Existing Run

```
conda activate MOF_sim
python benchmark.py benchmark_tao2022_mof5_co2.json --evaluate-all --run-id <run_id>
```

14 Appendix: Important Assumptions

- The current first benchmark system is CO₂ adsorption in MOF-5/IRMOF-1.
- The framework is treated as rigid.
- The current framework charge scheme is DDEC.
- CRAFTED is treated as an external raw-data source and is not committed as curated project data.
- The LAMMPS output molecule count is interpreted as absolute adsorption.
- Excess adsorption requires a pore volume and a bulk gas density model.
- The current pore volume for IRMOF-1 is taken from CRAFTED geometric descriptors.